Findings: soil study (evolved burrowers)

Oct 17, 2026 · @Nik

> **In short:** evolutionary suicide through the soil happens when burrowers evolve on their own. Burrowers took over, wore the soil down 17–47%, plants crashed 41–80%, and the burrowers died out, followed by everything else. A 30-vs-30 control shows the soil damage is the cause: with damage, all 11 burrower takeovers ended in extinction; with burrowing made harmless to the soil, all 5 survived to the end and their worlds lived on.
>
> On the baseline settings the pattern appeared in 10 of 30 runs. What usually stops it is the early predator wave, which wipes out the prey before claws can spread.

Answers to the project brief

The four presentation questions, answered from the soil study, with the evidence to show for each.

Brief question	Answer	Evidence to show
1. Did an extinction event occur?	**Yes.** Total extinction (prey, predators and plants) in all 111 sweep runs and 55 of 60 control runs. The only 5 worlds that survived were ones where claws spread and burrowing did no soil damage.	"Three ways a world ended" table; the claws-spread-vs-not chart

Brief question	Answer	Evidence to show
2. Mis-adaptation or failure to adapt?	**Both.** Failure to adapt in most runs: claws evolved but didn't spread before predators wiped out the prey (median tick 5,500). Mis-adaptation whenever burrowers took over and burrowing damaged the soil: the claw kept prey alive about 5× longer, then wore the soil down and the burrowers died out (10 of 30 baseline runs). The control confirms it was the trait's own soil damage that killed them.	Early-collapse chart (failure to adapt); 25_grid, 07 and control charts (mis-adaptation)
3. Environment vs the initial seeded species?	**The seeded species decided whether burrowers got a chance; the environment decided what happened to them.** 10 predators against 20 clawless prey caused the early wave in most runs, whatever the soil settings, even with burrowing blocked. In the control, with identical starting species, changing only the soil damage flipped the burrowers' fate from 11 of 11 dying to 0 of 5. The teammates' batch, with 10 burrowers seeded and free claws, agrees: there soil regen decided survival (fast 5/5, slow 0/5).	Control comparison chart; control 29 chart; comparison with the claw-soil-predator batch

Brief question	Answer	Evidence to show
4. Instrumental features and traits?	**Environment:** soil damage per burrow is decisive (11 of 11 burrower populations died with damage 5, 0 of 5 with damage 0, p ≈ 0.0002). Slow soil recovery (regen amount 0.5, regen 0.01%), rich starting ground (patch threshold −0.2), plants that need richer soil (min plant fertility 9) and the soil map (0 of 18 on map 1003) shaped the strongest cases. **Trait:** the burrowing claw, but only when the eye was kept (11 of 12 clawed prey in a replay were blind and never burrowed), and only with heavy use (over 1,700 burrows in every pattern run).	Control section; the 7-run table; "What blocks it"; map-1003 chart

Plants with killer cells evolved early in every run (by tick 200–1,700), so they were part of the background in all worlds. This study can't separate their effect from the soil's.

Setup

31 configurations × 3 runs on soil map 42 (93 runs), plus 6 of those configurations × 3 runs on soil map 1003 (18 runs). Each run went up to 150,000 ticks on a 200 × 150 grid, headless. The 3 runs of a configuration share the map and starting positions; only the random events differ.

Setting	Value in every run
Starting organisms	10 predators, 20 prey without claws, **0 prey with claws**, 39 plants
Burrowing	hides 50 ticks, removes 5 fertility per burrow, needs fertility ⩾ 1

Setting	Value in every run
When prey burrow	whenever they see a predator (burrow trigger range 1, stay hidden range 8, only-when-close off)
Claws	not free
Soil baseline	regen 0.05% chance per tick, regen amount 1, patch threshold 0 (~51% fertile), patch resolution 20, min plant fertility 7, max fertility 10

Each sweep changed one soil setting from the baseline. Four grid runs paired slow or fast regen with less or more fertile ground, and two controls blocked burrowing or made it harmless to the soil.

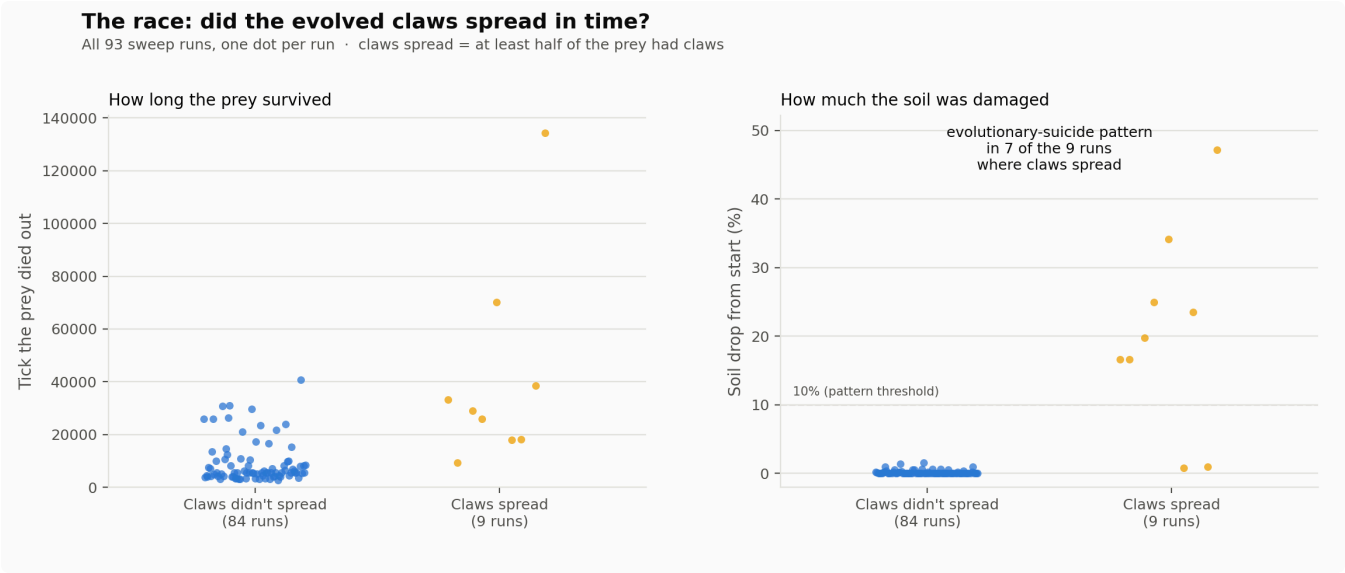
Three ways a world ended

All 111 runs ended in total extinction, but they got there in three different ways. Which way depended on whether claws spread, and then on how much the clawed prey burrowed.

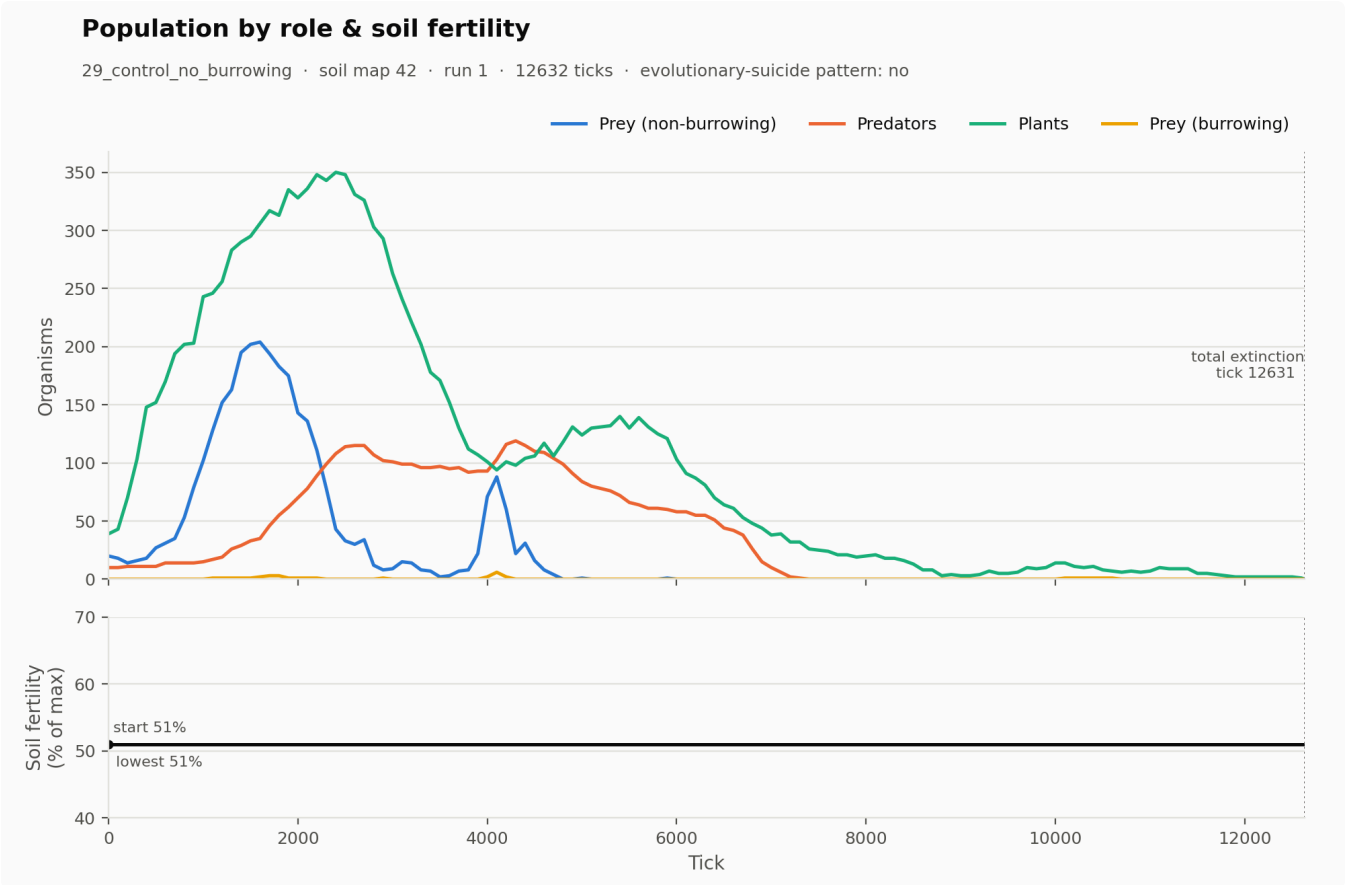
Outcome	Runs (map 42 / map 1003)	Prey gone (median tick, map 42)	Burrows per run	Soil drop
Early collapse: predators wiped out the prey before claws reached half of them	84 / 16	5,500	median 0–1, max 173	none
Claws spread, little digging: clawed prey took over but burrowed rarely	2 / 2	18,000 and 38,600	61–153	⩽ 1%
Evolutionary suicide: claws took over, burrowing wore down the soil, burrowers died out	7 / 0	28,900	1,877–8,872	17–47%

The early collapse is a failure to adapt, not a soil effect: it happened just as often when burrowing was blocked (control 29) or did no soil damage (control 30). Prey went first, then predators, then plants in 63 of 93 map-42 runs.

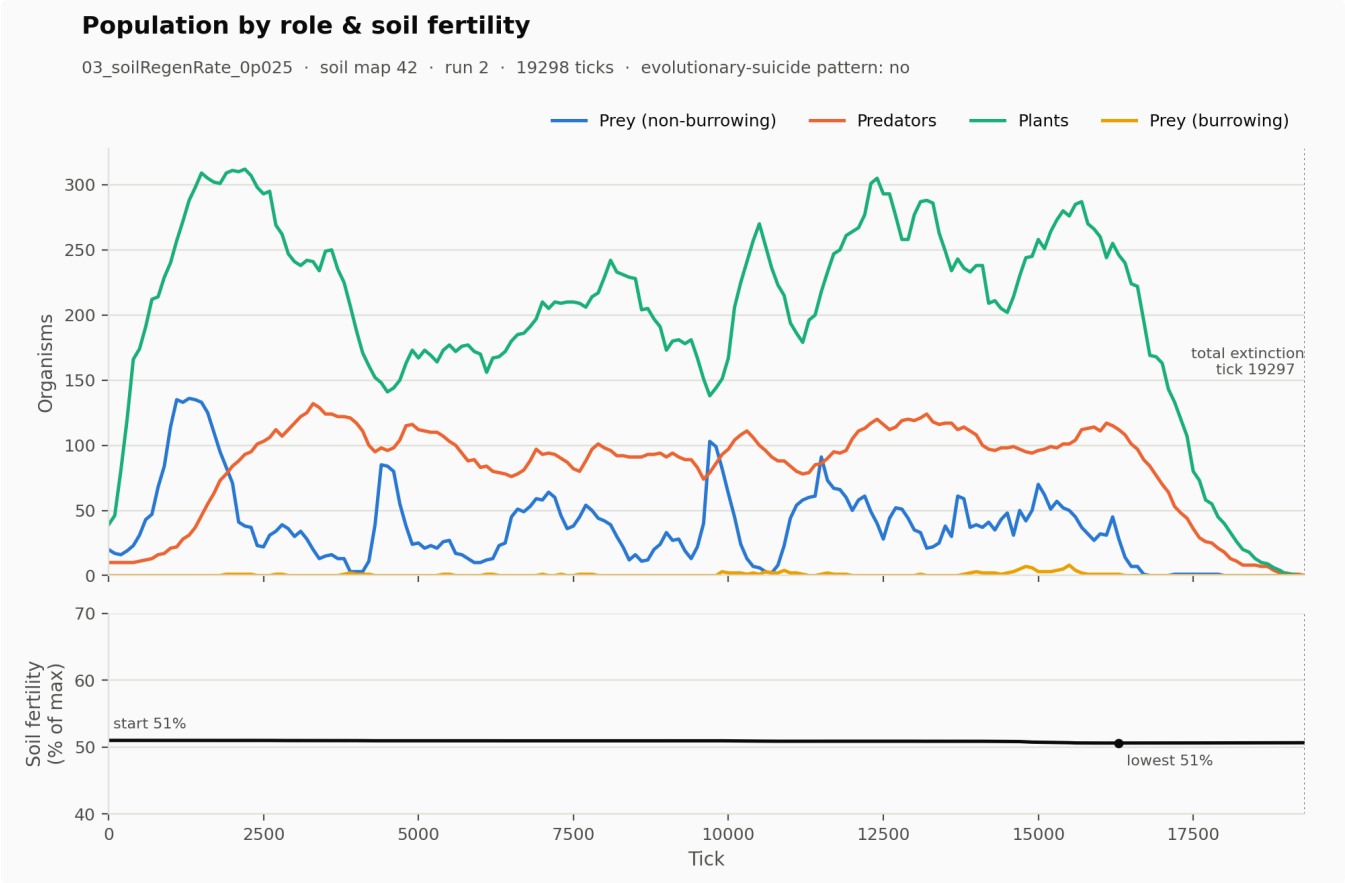
The line between the last two rows is the amount of digging. Every run with more than about 1,800 burrows wore the soil down and ended in the pattern; no other run, on either map, went above 173 burrows.



All 93 map-42 runs. Left: runs where claws spread kept prey alive far longer. Right: only those runs damaged the soil, and 7 of the 9 crossed the 10% pattern threshold.



Early collapse (29_control_no_burrowing, run 1). Prey are gone by tick ~4,800 with no burrowing and flat soil; the rest die out by tick 12,631.



Claws spread, little digging (03_soilRegenRate_0p025, run 2). Only 95 burrows, so the soil stays at 51% and plants keep cycling until the prey die out.

The 7 evolutionary-suicide runs

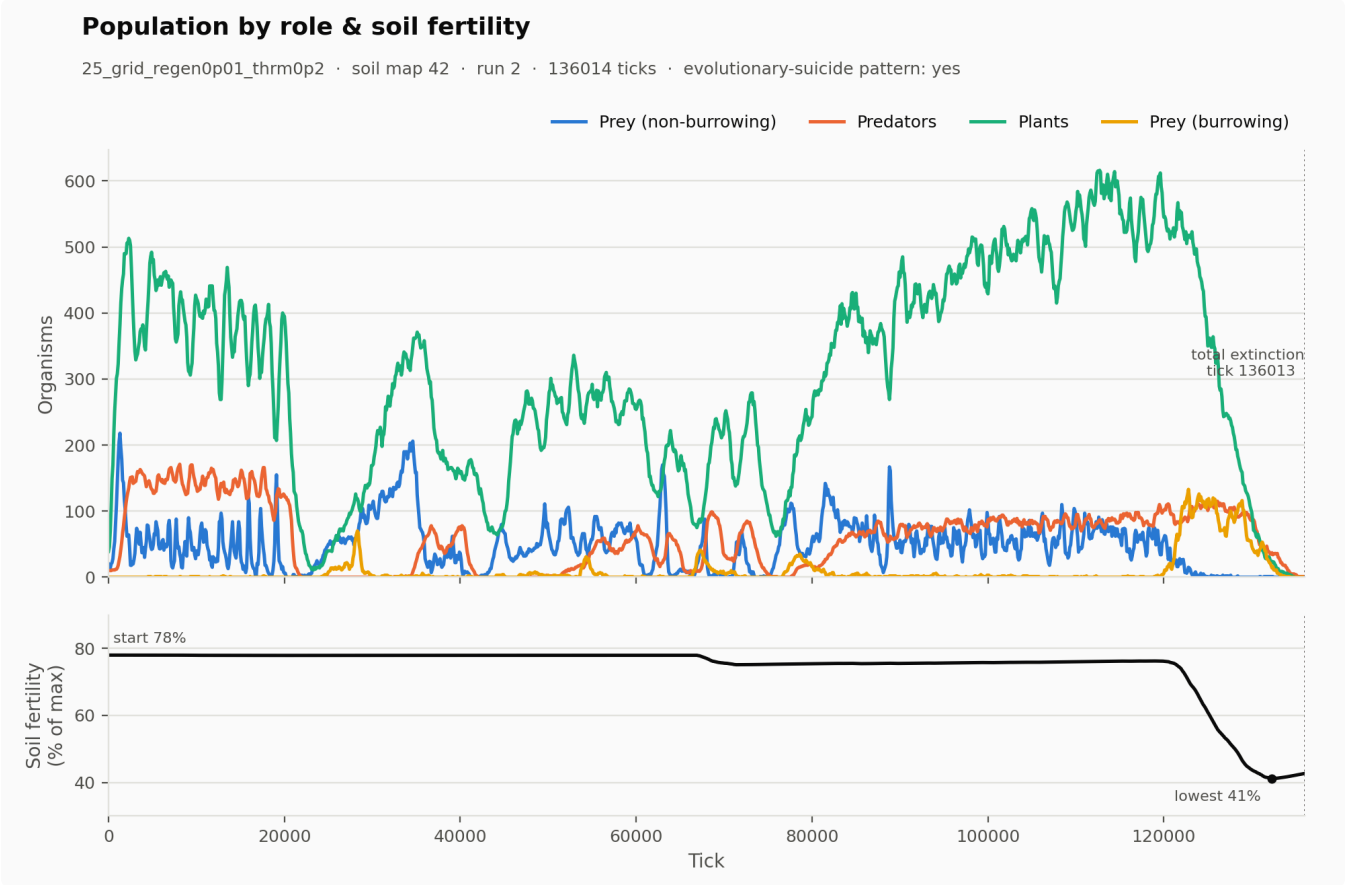
All 7 followed the same chain: claws took over, burrowing wore the soil down, plants thinned out, then the burrowers died out 2,100–9,000 ticks after the soil hit its lowest point. Everything else was gone 1,100–10,100 ticks after that.

Run (map 42)	Changed from baseline	Claws peak (tick)	Burrows	Soil fertility, start → lowest	Plants, before claws peaked → at soil low	Burrowers gone (tick)	All extinct (tick)
25_grid run 2	regen 0.01%, patch threshold −0.2 (78% fertile)	70,300 (main wave from ~121,000)	8,872	78% → 41%	156 → 64	134,400	136,013

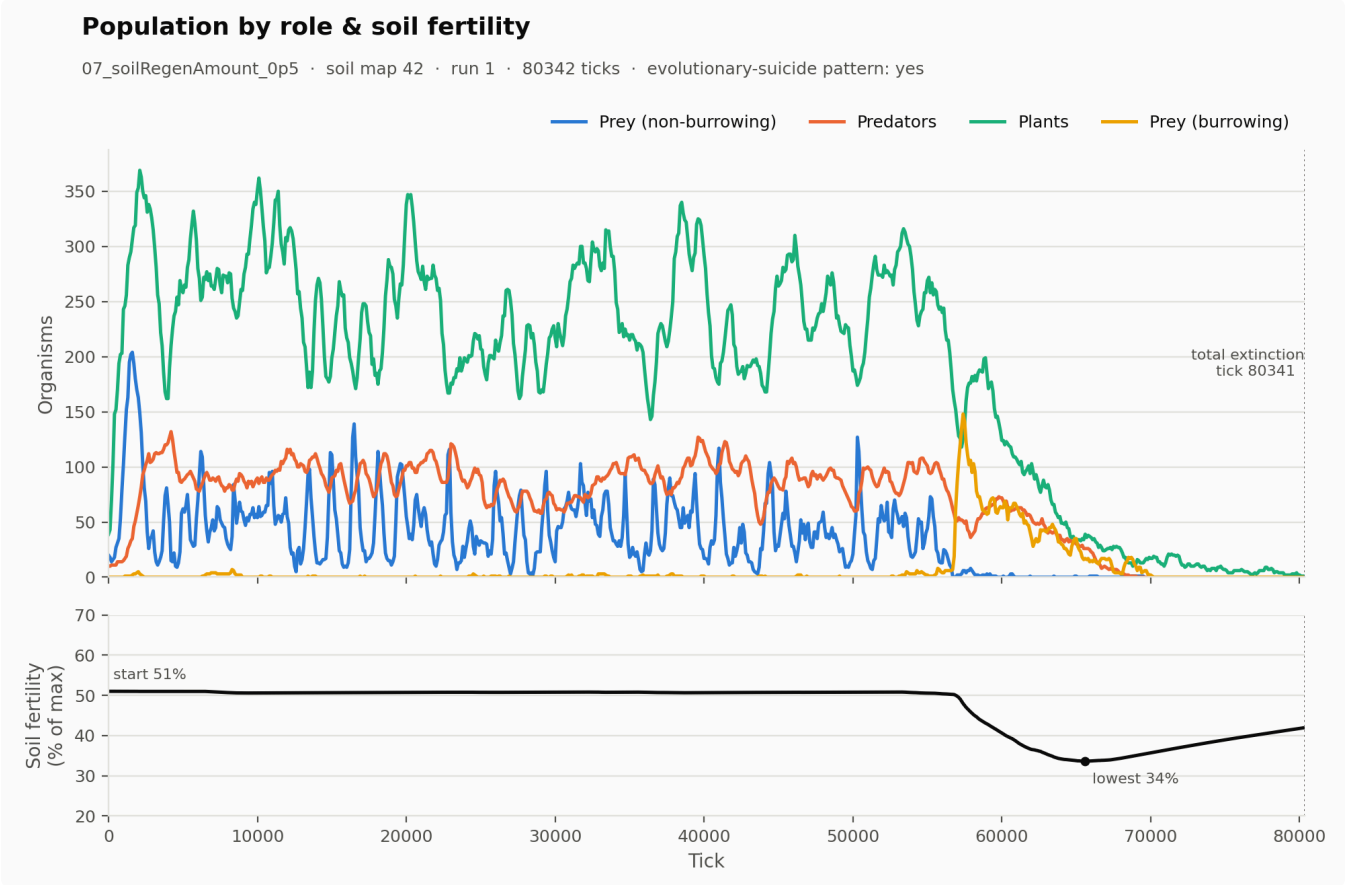
Run (map 42)	Changed from baseline	Claws peak (tick)	Burrows	Soil fertility, start → lowest	Plants, before claws peaked → at soil low	Burrowers gone (tick)	All extinct (tick)
07 run 1	regen amount 0.5	56,800	4,674	51% → 34%	262 → 54	70,200	80,341
23 run 2	soil map 7	18,800	4,167	52% → 39%	160 → 51	25,900	27,504
21 run 2	min burrow fertility 0	5,600	4,188	51% → 39%	218 → 44	18,200	19,261
05 run 2	regen 0.12%	16,100	4,891	51% → 41%	233 → 108	28,900	31,406
00 run 1	none (baseline)	18,100	3,057	51% → 43%	230 → 136	33,200	35,368
17 run 1	min plant fertility 9	4,600	1,877	51% → 43%	237 → 53	9,400	11,184

"Plants" is the average plant count over the 5,000 ticks before claws peaked against the 3,000 ticks before the soil's lowest point. Charts: `soil-tests/figures/runs/map42/<run>_pop_soil.png`.

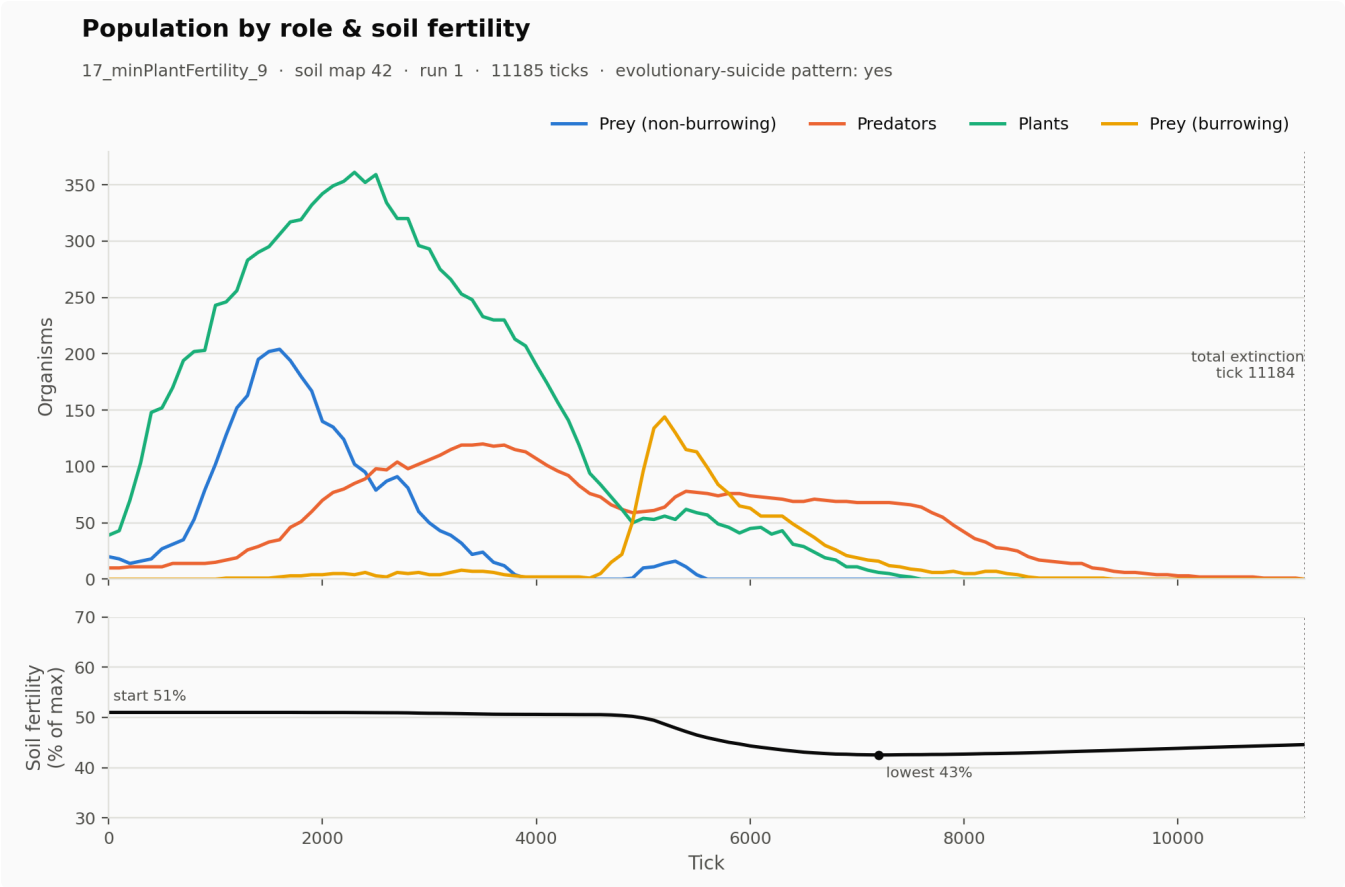
- **Plants crashed in every one:** down 41–80% between the claw peak and the soil low. In 3 runs (17, 21, 23) the plants died out before the burrowers did.
- **Soil recovered afterwards:** once the burrowers were gone, fertility climbed back to 43–50%, too late for them.
- **The clearest example is 25_grid run 2:** it lasted 136,013 ticks, the longest of any run, and had the biggest soil drop (37 percentage points).



25_grid run 2. Burrowers (yellow) surge from tick ~121,000, soil falls from 76% to 41%, plants crash from ~550 to zero, and everything is extinct by tick 136,013.



07_soilRegenAmount_0p5 run 1. Claws take over at tick ~57,000, soil drops from 50% to 34% over the next 8,000 ticks, plants thin out and the burrowers are gone by tick 70,200. Soil starts recovering only after they die.



17_minPlantFertility_9 run 1, the fastest chain. Plants need fertility 9, so a small soil drop removes them first; burrowers follow at tick 9,400.

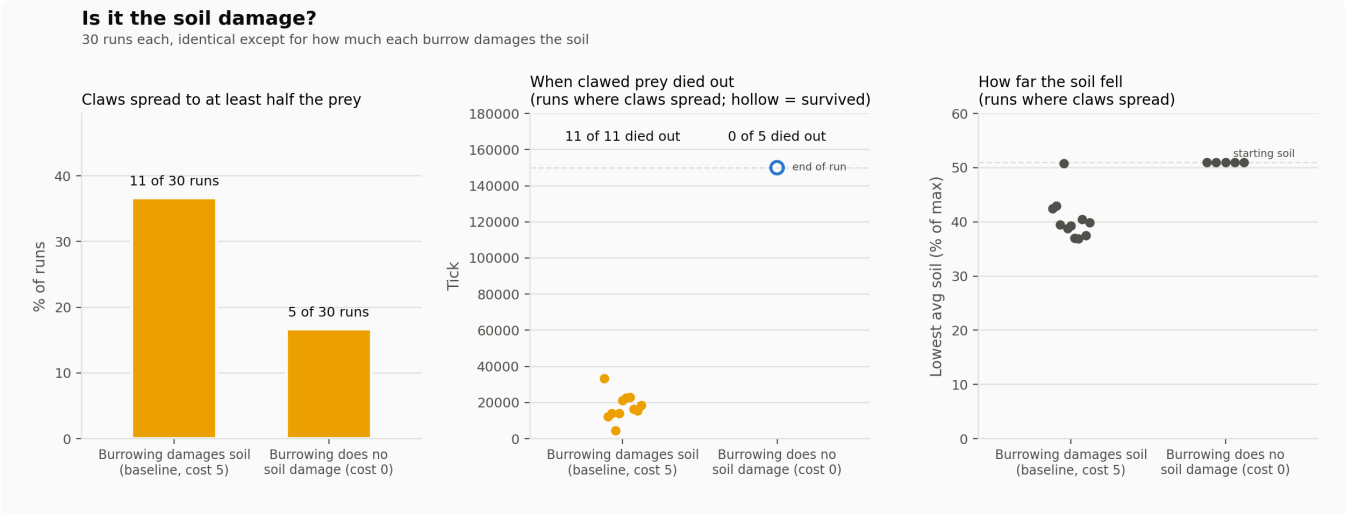
The control: soil damage is what kills the burrowers

When burrowing damaged the soil, every burrower takeover ended in the burrowers' extinction (11 of 11). When the same burrowing did no soil damage, every takeover survived to the end of the run (5 of 5), and so did its world. Both batches had 30 runs with the same map, starting species and random seeds; only the soil damage per burrow changed.

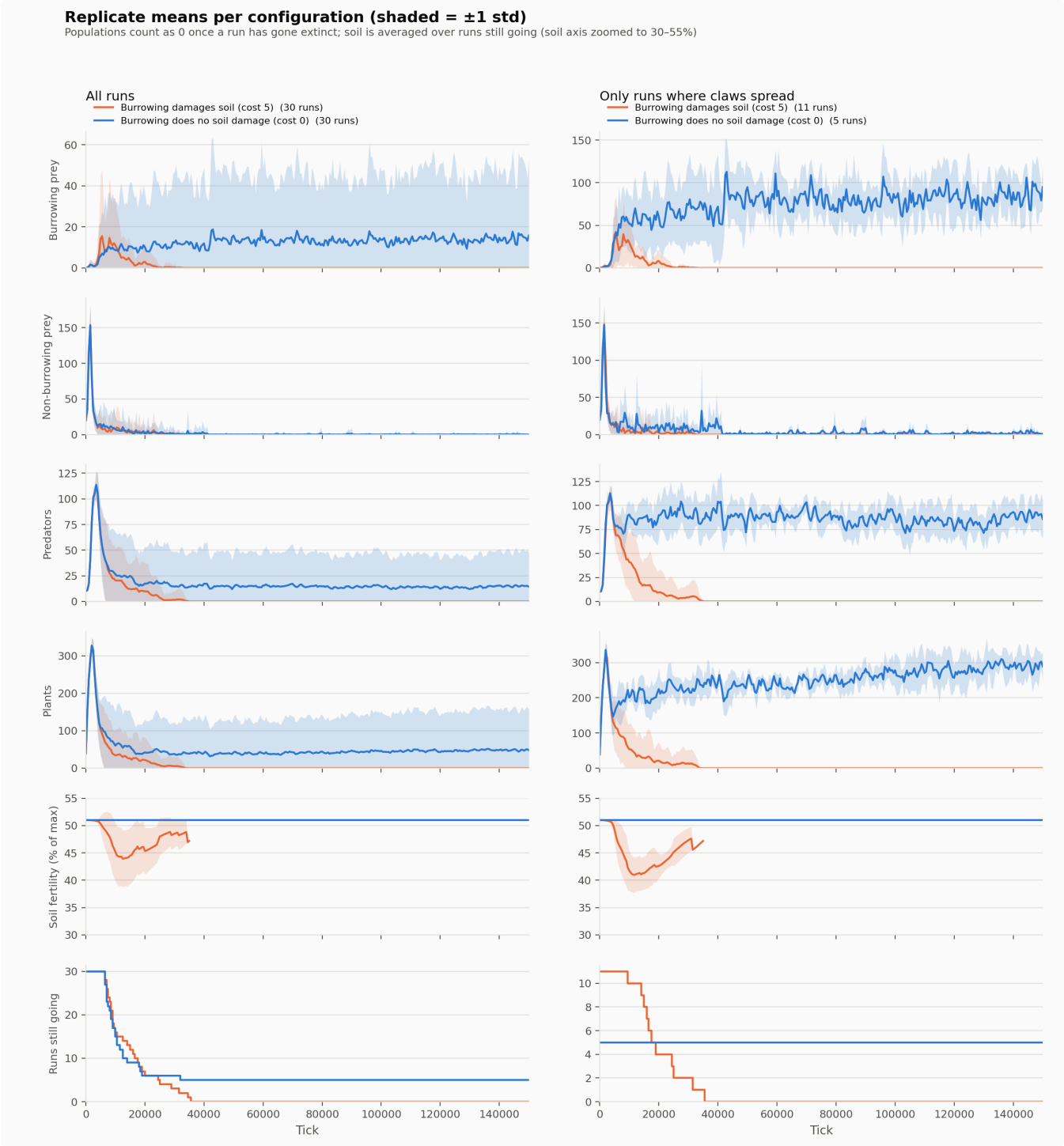
	Burrowing damages soil (cost 5, baseline)	Burrowing does no soil damage (cost 0)
Runs	30	30
Claws spread to half the prey	11	5

	Burrowing damages soil (cost 5, baseline)	Burrowing does no soil damage (cost 0)
Burrowers died out after spreading	**11 of 11**	**0 of 5**
Lowest soil in those runs	37–51% (median 39.5%)	51% (never moved)
Burrows in those runs	35–4,530	58,825–160,739
World alive at 150,000 ticks	0 of 30	5 of 30 (the same 5 where claws spread)
Evolutionary-suicide pattern	10 of 30	0 of 30

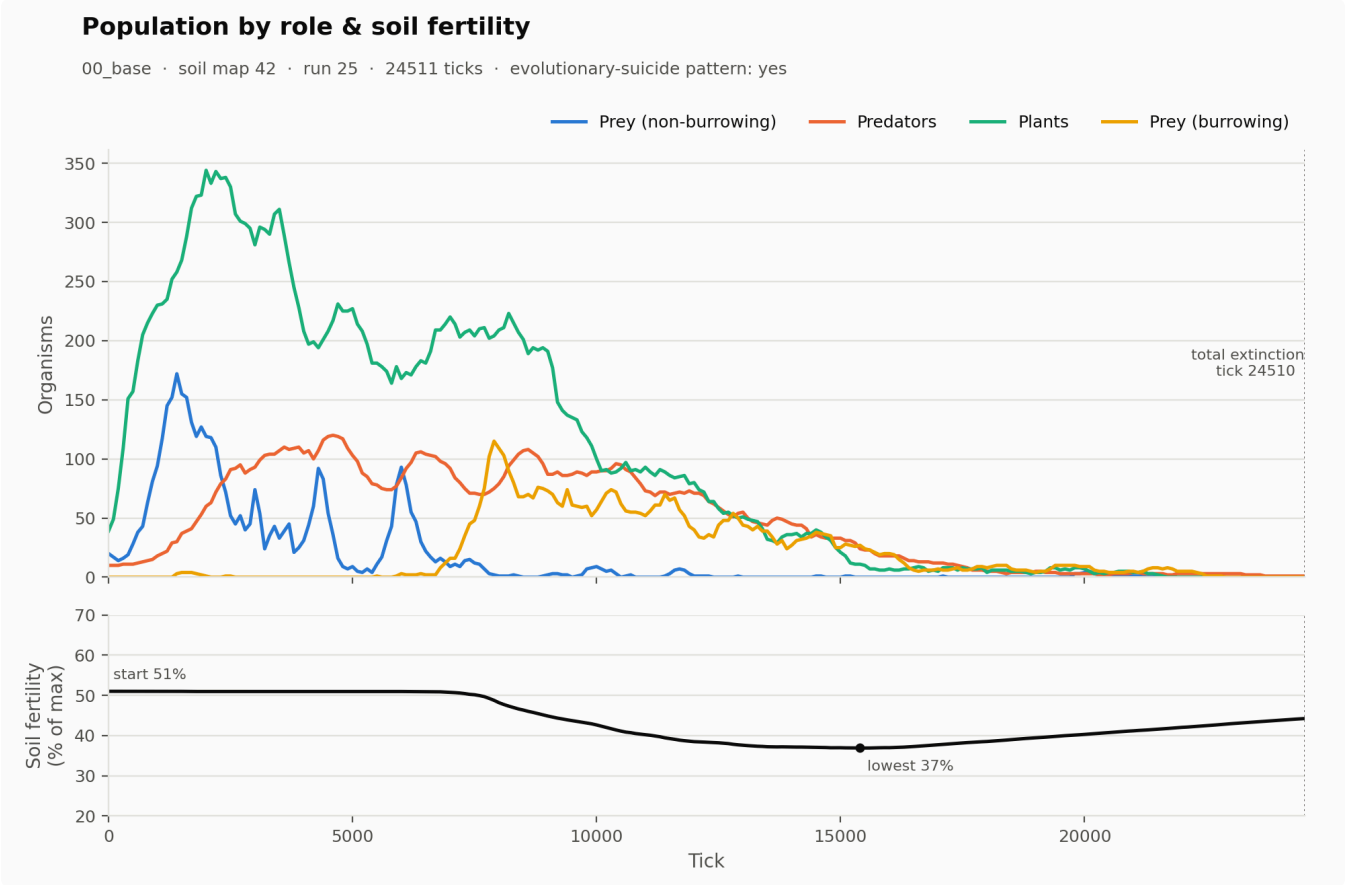
The chance of an 11-of-11 vs 0-of-5 split happening by luck is about 0.02% (Fisher's exact test, p ≈ 0.0002). The gap in how often claws spread (11 vs 5) is within chance. Without soil damage, burrowers dug 15–35× more and plants stayed at about 150–330 for the whole run. The 25 no-damage runs where claws didn't spread still went extinct from the early predator wave.



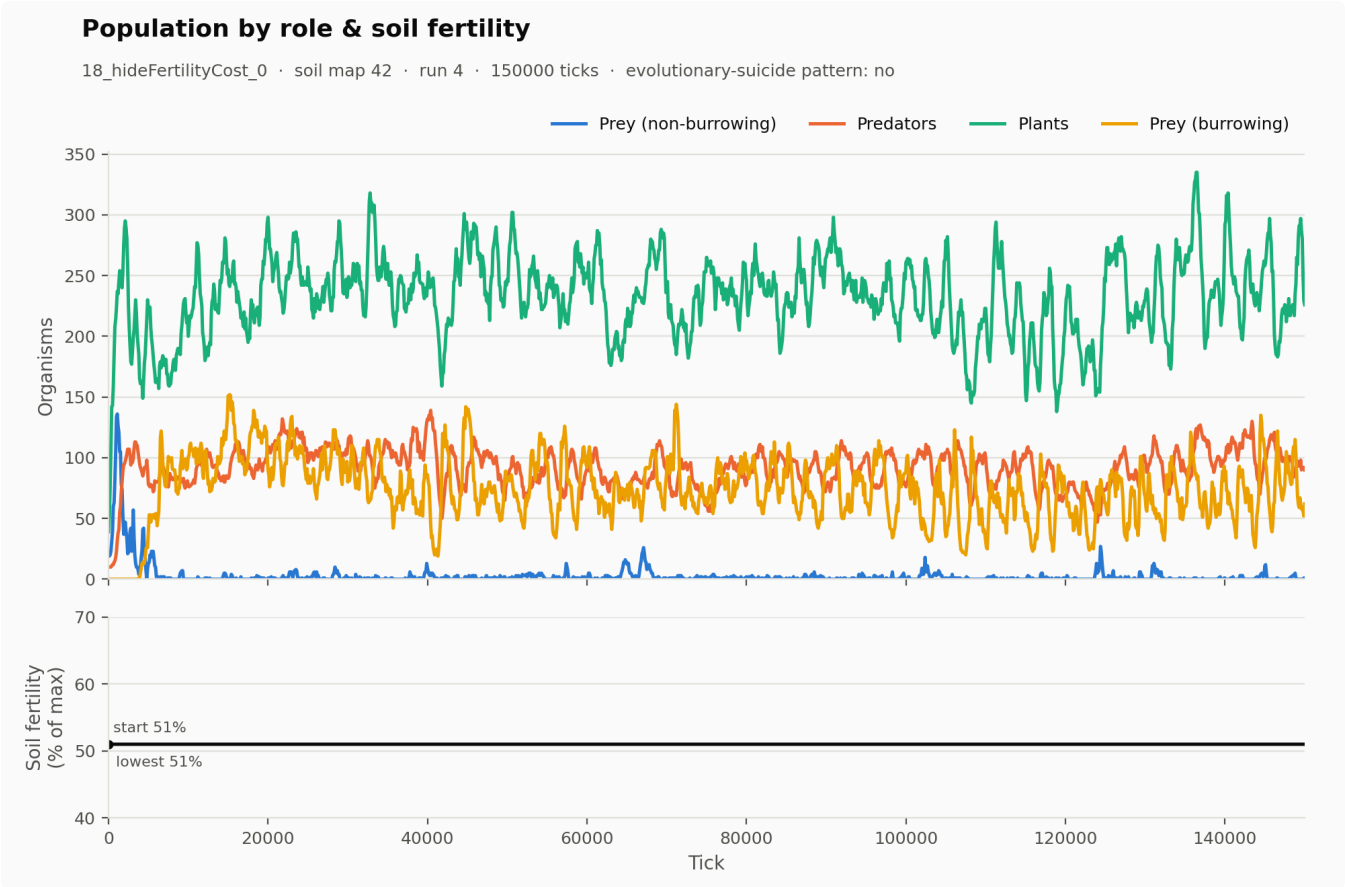
Left: how often claws spread. Middle: every clawed population died out when burrowing damaged the soil, none when it didn't (hollow = survived to tick 150,000). Right: soil fell to 37–43% only when burrowing damaged it.



Replicate means (±1 std) over time. Right column, runs where claws spread: with soil damage (orange) the soil dips to about 41%, plants and predators fall away and all 11 burrower populations are gone by tick ~35,000. Without damage (blue) the soil stays flat at 51%, plants climb to about 300, and burrowers, predators and plants all persist to 150,000 ticks. Left column, all runs: both batches lose most worlds to the early predator wave at the same rate.



With soil damage (baseline, run 25): burrowers take over, soil falls from 51% to 37%, plants thin out and the burrowers are gone by tick 22,900.



Without soil damage (cost 0, run 4): burrowers take over by tick ~5,000 and live alongside predators and plants until the run ends at 150,000 ticks. The soil never moves.

Data: `soil-tests/results_control/`; chart: `soil-tests/figures/control_comparison.png`.

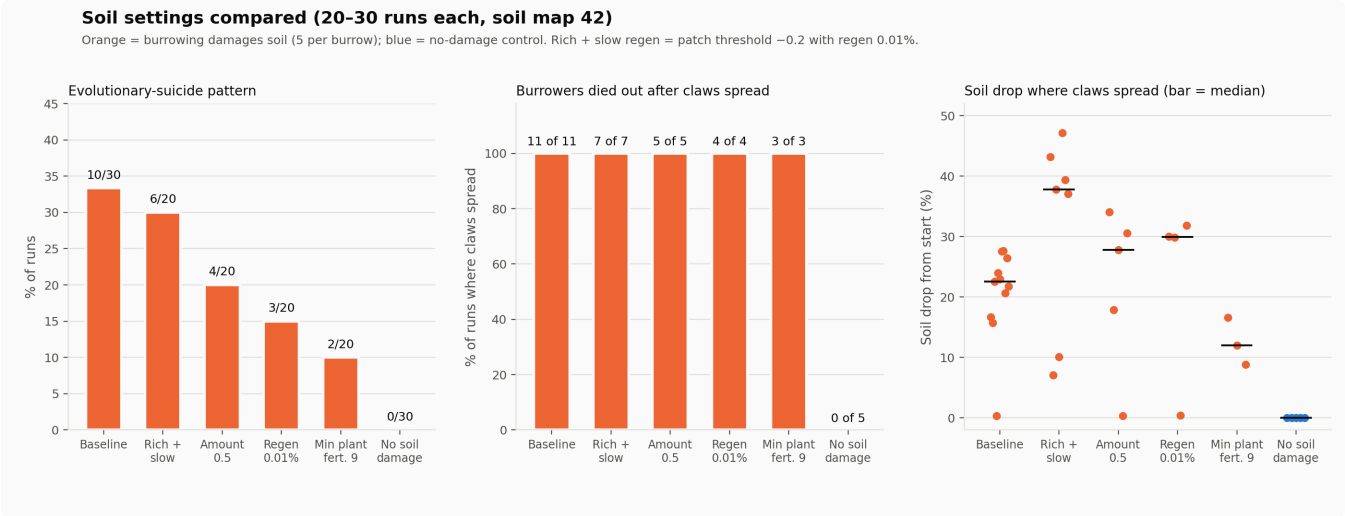
Best parameters

Rerun with 20–30 runs each, no soil setting beat the baseline at producing the pattern. The soil settings change how hard the soil gets hit, not whether the burrowers die: once claws spread with soil damage on, the burrowers died out in 30 of 30 takeovers across all five settings, against 0 of 5 with no damage ($p \approx 0.000003$).

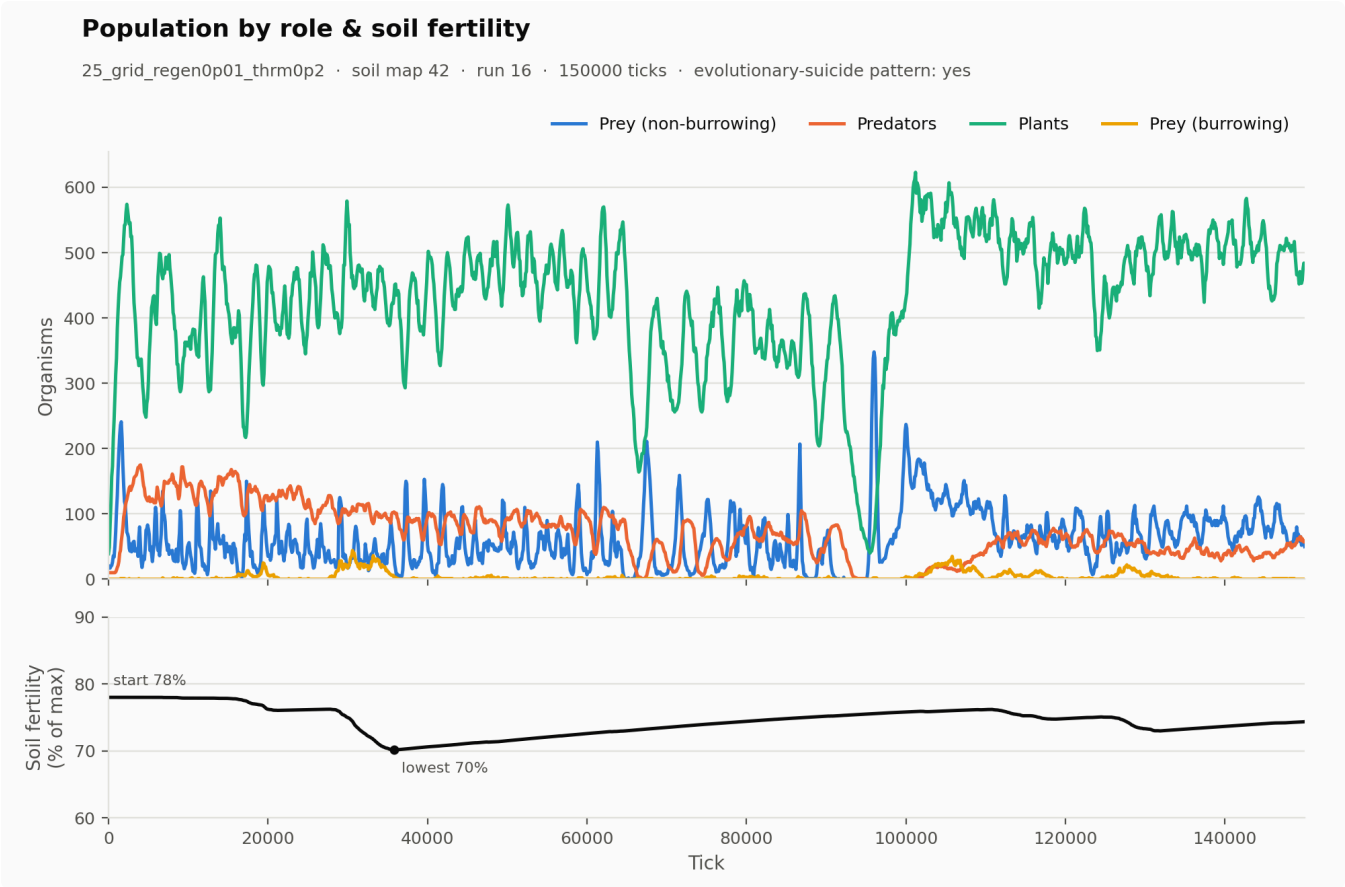
Setting (everything else baseline)	Runs	Evolutionary-suicide pattern	Burrowers died once claws spread	Median soil drop where claws spread
Baseline (regen 0.05%, amount 1, threshold 0)	30	**10 (33%)**	11 of 11	23%

Setting (everything else baseline)	Runs	Evolutionary-suicide pattern	Burrowers died once claws spread	Median soil drop where claws spread
Rich ground + slow regen (threshold −0.2, regen 0.01%)	20	6 (30%)	7 of 7	**38%**
Regen amount 0.5	20	4 (20%)	5 of 5	28%
Regen 0.01%	20	3 (15%)	4 of 4	30%
Min plant fertility 9	20	2 (10%)	3 of 3	12%
No soil damage (control)	30	0	0 of 5	0%

The differences in pattern rate are within chance ($p \geq 0.19$ against the baseline, Fisher's exact test): how often the pattern appears is set by whether claws spread, which the predator wave decides. Slow soil recovery deepens the damage (30–38% drops against 23%). Min plant fertility 9 shows the smallest drops because plants die off at the first small dip, so burrowers starve sooner.



Left: how often each setting showed the pattern. Middle: burrowers died out in every takeover whenever burrowing damaged the soil, never without damage. Right: how far the soil fell; slow recovery gives the deepest drops.



Rich ground + slow regen, run 16: the only world with soil damage that survived. The burrowing lineage dug the soil from 78% to 70% and died out, while non-burrowing prey, predators and plants carried on to 150,000 ticks. The burrowers destroyed only themselves.

Settings where claws never spread in any of their 3 runs, so they cannot show the pattern:

- Patch threshold on its own (±0.1, ±0.2): all collapsed by tick 8,700 with almost no burrowing.
- Patch resolution 10, 40 or 60.
- Regen amount 2 (soil heals too fast), min plant fertility 5, min burrow fertility 3, soil map 99.
- Soil damage per burrow 0, 2 or 10. Only the baseline 5 ever showed the pattern; 0 is the no-damage control.

Best parameters to present: the baseline for how often it happens (10 of 30 runs), and rich ground + slow regen (patch threshold −0.2, regen 0.01%) for the clearest soil signal (median 38% drop, up to 47%). Data: `soil-tests/results_control/soil_settings_compared.csv`; chart: `soil-tests/figures/soil_settings_compared.png`.

What blocks it

Two things stand in the way, and neither is a soil setting.

1. **The predator wave comes first.** Claws appeared in 87 of 93 runs (median tick 1,200), but prey were usually gone by tick 5,500, while claws were still about 4% of the prey.
2. **Most claw mutations make the prey blind.** In a replayed run, 11 of 12 clawed prey had lost their eye to the claw. A prey only burrows when it sees a predator, so a blind clawed prey never digs and gets no protection.

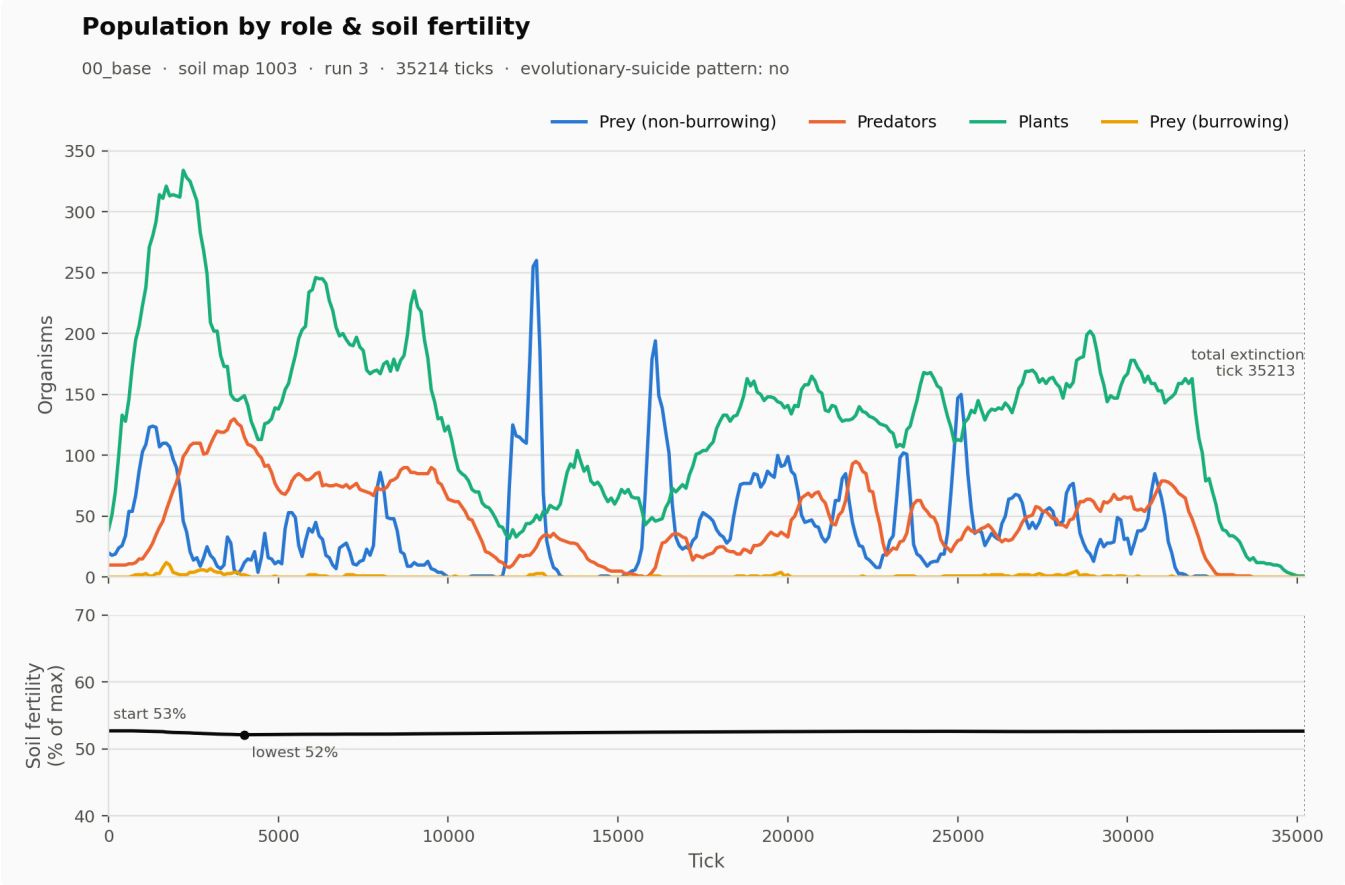
The burrowing itself works as intended: in a one-prey, one-predator test, the clawed prey burrowed on sight, each burrow removed exactly 5 fertility, it couldn't be eaten underground and resurfaced after 50 ticks (`soil-tests/check_mechanics.js`, 6 of 6 checks passed).

Second soil map (1003)

The 6 configurations that showed the pattern on map 42 were rerun on map 1003 (53% fertile), 3 runs each. None of the 18 showed the pattern.

	Map 42	Map 1003
Runs	93	18
All extinct	93	18
Claws spread to half the prey	9	2
Evolutionary suicide	7	0

In the 2 map-1003 runs where claws spread, the prey burrowed only 61 and 153 times and the soil moved by under 1 percentage point. The rich-ground + slow-regen run collapsed early in all 3 tries (median tick 8,152). The pattern depends on the map and on chance, not on any one soil setting.



Map 1003, 00_base run 3. Claws reach 57% of prey, but with only 153 burrows the soil never moves, and the burrowers die out anyway.

How this compares with the claw-soil-predator batch

The two batches answer different questions, so they don't contradict each other. That batch asks what decides survival once burrowers exist; this one asks whether burrowing can evolve on its own and then cause its own extinction.

	Claw-soil-predator batch	This soil study
Prey with claws at the start	10	0 (claws must evolve)
Prey without claws at the start	10	20
Claws free	yes	no
Grid	384 × 193, cell size 5	200 × 150, cell size 4

	Claw-soil-predator batch	This soil study
Runs	6 configurations × 5, up to 250,000 ticks	37 configurations, 111 runs, up to 150,000 ticks
Soil regen tested (% chance per tick)	0.01, 0.05, 0.2	0.005 to 0.2
Worlds that survived	fast regen 5/5, baseline 3/5, slow regen 0/5	0 of 111

With burrowers present from tick 0 and free claws, prey hide straight away and survive the first predator wave, so the soil then decides survival. Here the burrowers must evolve first, and predators usually win before the soil matters. Both agree that once burrowing is common, it wears the soil down.

Caveats and next steps

- **Small numbers for the sweeps:** the full sweep had 3 runs per setting. The 4 most promising settings were rerun 20 times and the baseline 30 times, so those comparisons hold; the other sweep settings are still a lead, not proof.
- **Cause shown for soil damage only:** the control proves soil damage is what kills the burrowers. It doesn't separate which downstream path matters most (fewer plants to eat, or less fertile ground to burrow into).
- **The predator wave:** most worlds die before claws spread, even with burrowing blocked. A starting balance that survives without burrowing (fewer or slower predators, a teammate's area) would make the pattern more common.
- **Grid size:** headless runs use a smaller grid than the browser, so soil map 42 here is not identical to the browser's map 42. The map generator also stretches patches by columns ÷ rows (about 1.3× taller than wide here, 2× in the browser), a quirk inherited from the original LifeEngine that is the same in every run.